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Chapter II.6, Week 8

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6.1. Let X be a scheme satisfying $(*)$. Then $X \times \mathbb{P}^n$ also satisfies $(*)$, and $\mathrm{Cl}(X \times \mathbb{P}^n) \cong \mathrm{Cl}(X) \times \mathbb{Z}$.

Proof. We note that $X \times \mathbb{P}^n$ is covered by finitely many $X \times U_i \cong X \times \mathbb{A}^n$. $X \times \mathbb{A}^n$ satisfies $(*)$ by (6.6) and induction on n. It follows that $X \times \mathbb{P}^n$ is noetherian, reduced, and regular in codimension one. Each $(X \times U_i) \cap (X \times U_j) = X \times (U_i \cap U_j)$ is nonempty, so $X \times \mathbb{P}^n$ is irreducible, and hence integral. It suffices, then, to show that $X \times \mathbb{P}^n$ is separated.

Let $\mathrm{pr}_2 : X \times \mathbb{P}^n \to \mathbb{P}^n$ denote the projection. Note that our cover $\{U_i\}$ has $\mathrm{pr}_2^{-1}(U_i) = X \times U_i$. The maps $X \times U_i \to U_i \to \operatorname{Spec} \mathbb{Z}$ are separated, so that $X \times U_i \to U_i$ is separated. It follows that $X \times \mathbb{P}^n \to \mathbb{P}^n$ is separated, so $\mathbb{P}^n \to \operatorname{Spec} \mathbb{Z}$ separated gives us that $X \times \mathbb{P}^n$ is separated.

Consider the closed subset $H := X \times V(x_0) \subset X \times \mathbb{P}^n$. It is easily seen to be irreducible of codimension one, so that (6.5) gives the following exact sequence:

$$\mathbb{Z} \to \mathrm{Cl}(X \times \mathbb{P}^n) \to \mathrm{Cl}(X \times U_0) \to 0.$$

(6.6) and induction on n once again gives us $\mathrm{Cl}(X \times U_0) \cong \mathrm{Cl}(X)$. We begin by showing $\mathbb{Z} \to \mathrm{Cl}(X \times \mathbb{P}^n)$ is injective. We will first show that $H \in \mathrm{CaCl}(X \times \mathbb{P}^n) \cong \mathrm{Pic}(X \times \mathbb{P}^n)$, then show $\mathscr{L}(H) \mapsto 1 \in \mathbb{Z}$ under a series of maps.

For any open affine $\operatorname{Spec} A \cong V \subset X$, $V \times \mathbb{P}^n \cong \mathbb{P}^n_A$, and $H|_{V \times U_i} = (x_0)$ for all i, j with $j \neq 0$. It follows that $H \in \mathrm{CaCl}(X \times \mathbb{P}^n) \cong \mathrm{Pic}(X \times \mathbb{P}^n)$, and $\mathscr{L}(H)|_{V \times \mathbb{P}^n} \cong \mathscr{O}_{\mathbb{P}^n_A}(1)$. Now fix a particular $\operatorname{Spec} A \cong V \subset X$, and a point $x \in V$. We can base extend $\operatorname{Spec} k(x) \to \operatorname{Spec} A$ to a map $\mathbb{P}^n_{k(x)} \to \mathbb{P}^n_A$. Composing with $\mathbb{P}^n_A \to X \times \mathbb{P}^n$ gives us a map $f : \mathbb{P}^n_{k(x)} \to X \times \mathbb{P}^n$. (Ex. 6.8) gives us a map $f^* : \mathrm{Pic}(X \times \mathbb{P}^n) \to \mathrm{Pic}(\mathbb{P}^n_{k(x)}) \cong \mathbb{Z}$ such that

$$\mathscr{L}(H) \mapsto \mathscr{O}^n_{k(x)} \mapsto 1 \in \mathbb{Z}.$$

Then $m \cdot H \mapsto m \in \mathbb{Z}$ under this series of maps, so that in particular $m \cdot H \neq 0$. Hence, $\mathbb{Z} \to \mathrm{Cl}(X \times \mathbb{P}^n)$ is injective. Note as well that the above defines a map $\mathrm{Cl}(X \times \mathbb{P}^n) \to \mathrm{Cl}(\mathbb{P}^n_{k(x)}) \cong \mathbb{Z}$ such that the following diagram, which we have just shown is exact, commutes.

$$\begin{tikzcd}
0 \arrow[r] & \mathbb{Z} \arrow[r] & \mathrm{Cl}(X \times \mathbb{P}^n) \arrow[r] \arrow[d] & \mathrm{Cl}(X) \arrow[r] & 0 \\
& & \mathrm{Cl}(\mathbb{P}^n_{k(x)}) \arrow[ul, "\cong"] & &
\end{tikzcd}$$

The splitting lemma then implies that, in fact, $\mathrm{Cl}(X \times \mathbb{P}^n) \cong \mathrm{Cl}(X) \times \mathbb{Z}$. □

6.5. *Quadric Hypersurfaces.* Let $\operatorname{char} k$, and let X be the affine quadric hypersurface $\operatorname{Spec} k[x_0, \dots, x_n]/(x_0^2 + x_1^2 + \cdots + x_r^2)$—cf. (I, Ex. 5.12).

(a) Show that X is normal if $r \geqslant 2$ (use (Ex. 6.4)).

(b) Show by a suitable linear change of coordinates that the equation of X could be written as $x_0 x_1 = x_2^2 + \cdots + x_r^2$. Now, imitate the method of (6.5.2) to show that:

(1) If $r = 2$, then $\operatorname{Cl} X \cong \mathbb{Z}/2\mathbb{Z}$;
(2) If $r = 3$, then $\operatorname{Cl} X \cong \mathbb{Z}$ (use (6.6.1) and (Ex. 6.3) above);
(3) If $r \geqslant 4$, then $\operatorname{Cl} X = 0$.

Proof. (a) Since $r \geqslant 1$, $f := x_1^2 + \cdots + x_r^2$ is non-constant. We will now show that f is square-free. Suppose $f = g^2$ for some $g \in k[x_1, \ldots, x_r]$. f is of degree 2, so $g = a + \sum_i a_i x_i$ for some $a, a_i \in k$. Then $f = g^2$ implies $a^2 = 0$, and $2a_i a_j = 0$ for all $i \neq j$. $\operatorname{char} k \neq 2$ implies $a_i a_j = 0$ for all $i \neq j$, so that at most one of the a_i is nonzero. Then $f = a_i^2 x_i^2$ implies f is a monomial, i.e. $r = 1$ and $f = x_1^2$. This contradicts $r \geqslant 2$, so f is indeed square-free.

Clearly $-f$ is non-constant and square-free as well. Thus, we may apply (Ex. 6.4) to show

$$\begin{aligned} A &= \frac{k[x_0, \ldots, x_r]}{(x_0^2 + x_1^2 + \cdots + x_r^2)} \\ &= \frac{k[x_1, \ldots, x_r][x_0]}{(x_0^2 - (-f))} \end{aligned}$$

is integrally closed, and hence normal. Then X is the spectrum of a normal ring, thus normal itself.

(b) The map given by $x_0 \mapsto \frac{i}{2}(y_0 + y_1)$, $x_1 \mapsto \frac{1}{2}(y_0 - y_1)$, and $x_i \mapsto y_i$ for all $1 < i \leqslant r$ is our desired change of coordinates. However, note that it is only well defined if $t^2 + 1$ splits in k, as x_0 maps to an element of $k(i)[y_0, y_1]$.

Clearly X is noetherian and seperated. If $r \geqslant 2$, then (I,5.12b) gives us X integral and (a) gives us X normal, whence regular in codimension 1. If $r = 2$, then $X = \operatorname{Spec} k[x, y, z]/(xy - z^2)$, so (1) is exactly (6.5.2).

Next, let $r = 3$. The same change of coordinates allows us to write $A = k[x, y, z, w]/(xy - zw)$, so that X is the cone of the surface Q defined in (6.6.1). Fix the hyperplane $H := V(x) \subset \mathbb{P}_k^3$, which does not contain Q. (Ex. 6.3b) gives the exact sequence

$$0 \to \mathbb{Z} \to \operatorname{Cl} Q \to \operatorname{Cl} X \to 0,$$

where $1 \mapsto Q.H \in \operatorname{Cl} Q$. $H \cap Q$ corresponds to the hyperplane $\{*\} \times \mathbb{P}^1$ under the isomorphism $Q \cong \mathbb{P}^1 \times \mathbb{P}^1$, so $Q.H \mapsto (1, 0)$ under $\operatorname{Cl} Q \cong \mathbb{Z} \oplus \mathbb{Z}$. Thus,

$$\operatorname{Cl} X \cong \frac{\mathbb{Z} \oplus \mathbb{Z}}{(1,0)\mathbb{Z}} \cong \mathbb{Z}$$

proves (3).

Finally, consider the case $r \geqslant 4$. Use our new coordinates to write $A = k[x_0, \ldots, x_r]/(x_2^2 + \cdots + x_r^2 - x_0 x_1)$, and consider the closed subscheme

$$Y = \operatorname{Spec} A/(x_0) \cong \operatorname{Spec} k[x_1, x_2, \ldots, x_r]/(x_2^2 + \cdots + x_r^2)$$

of X. $r - 2 \geqslant 2$, so $x_2^2 + \cdots + x_r^2$ is irreducible in $k[x_2, \ldots, x_r]$. Then

$$A/(x_0) \cong (k[x_2, \ldots, x_r]/(x_2^2 + \cdots + x_r^2))[x_1]$$

is an integral domain, so $x_2^2 + \cdots x_r^2$ is irreducible in $k[x_1, \ldots, x_r]$. Recalling that $A \cong k[x_0, \ldots, x_r]/(x_0^2 + \cdots + x_r^2)$ is quotient by an irreducible ideal as well, Krull's Hauptidealsatz gives us that $\dim A = r$ and $\dim(A/(x_0)) = r - 1$. Since $A/(x_0)$ is integral, $(x_0) \trianglelefteq A$ is prime, so (1.8A) gives us $\operatorname{codim}_X Y = \operatorname{ht}(x_0) = 1$. Hence, Y is a prime divisor of X.

By the above, (6.5) supplies us with the exact sequence

$$\mathbb{Z} \to \operatorname{Cl} X \to \operatorname{Cl}(X - Y) \to 0,$$

where $1 \mapsto 1 \cdot Y$. Note that (x_0) is the generic point of Y, so the local ring of Y is given by $\mathcal{O}_{X,(x_0)}$. The uniformizer is $\mathcal{O}_{X,(x_0)}(x_0)$, so clearly $v_Y(x_0) = 1$. Clearly $v_{Y'}(x_0) = 0$ for all other prime divisors Y', so $Y = (x_0)$ is principle. Thus, the exact sequence gives an isomorphism $\operatorname{Cl} X \cong \operatorname{Cl}(X - Y) = \operatorname{Cl} D(x_0)$. Recall that $D(x_0) \cong \operatorname{Spec} A_{x_0}$. We can write

$$\begin{aligned} A_{x_0} &= \frac{k[x_0, x_0^{-1}, x_1, x_2, \ldots, x_r]}{x_2^2 + \cdots + x_r^2 - x_0 x_1} \\ &\cong \frac{k[x_0, x_0^{-1}, x_1, x_2, \ldots, x_r]}{x_0^{-1}(x_2^2 + \cdots + x_r^2) - x_0} \\ &\cong k[x_0, x_0^{-1}, x_2, x_3, \ldots, x_r]. \end{aligned}$$

This is clearly a UFD, so we have that

$$\operatorname{Cl} X \cong \operatorname{Cl} A_{x_0} = 0.$$

This proves (4).

$\square$